

JEE MAIN REVISION MASTER: NUCLEAR CHEMISTRY 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 1 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Kcet Nuclear Chemistry - Core Application	1 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Nuclear Chemistry - Core Application (Standard Yield — 1 PYQs)

Core Principle: Key problem pattern in Kcet Nuclear Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

• [2024 Shift] Q#1 (MCQ): "The synthetically produced radioactive noble gas by the collision of ${}_{98}^{249}\text{Cf}$ with ${}_{20}^{48}\text{Ca}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The synthetically produced radioactive noble gas by the collision of ${}_{98}^{249}\text{Cf}$ with ${}_{20}^{48}\text{Ca}$ is

- (A) radon
- (B) radium
- (C) oganesson
- (D) xenon

Correct Answer: None

Solution Analysis: The reaction of ${}_{98}^{249}\text{Cf}$ and ${}_{20}^{40}\text{Ca}$ takes place as follows ${}_{98}^{249}\text{Cf} + {}_{20}^{40}\text{Ca} \rightarrow {}_{118}^{294}\text{Og} + 3{}_0^1\text{n}$. Thus, the radioactive noble gas formed is Oganesson (Z=118)